

OPERATIONALIZED INTEGRITY PRINCIPLES (OIP) v1.1

A Framework for Epistemic Governance of Large Language Model Outputs in High-Stakes Domains

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OVERVIEW

This archive contains the complete Operationalized Integrity Principles (OIP) framework, developed to address systematic epistemic failures in large language model outputs for high-stakes domains including medical, legal, financial, and regulatory analysis.

OIP operationalizes key principles from epistemological philosophy into mechanical safeguards against common LLM failure modes: for example citation laundering, false precision, narrative closure, and abstraction reification.

FRAMEWORK COMPONENTS

OIP comprises three integrated layers:

1. OIP-Core: Lightweight behavioral constraints (<1000 tokens) for baseline reasoning quality
2. OIP-JSON: Structured enforcement with regulatory mappings (NIST AI RMF, EU AI Act, OECD) – this layer includes a standard schema and a persistent enforcement schema for multi-turn deployment
3. OIP-TF: Comprehensive testing framework with Normalized Integrity Scoring (NIS)

ARCHIVE CONTENTS

Main Documents:

- OIP_Academic_Paper.pdf - Full research paper (38 pages + references)

Schema and Other Files:

- OIP Handbook
- OIP-Core_Release_1.1 - Core behavioural constraints specification (prompt injection)
- OIP-JSON_Standard_Release_1.1.json - Standard enforcement layer
- OIP-JSON_Persistent_Release_1.1.json - Session-persistent variant
- OIP-TF_Release_1.1.json - Testing framework with evaluation protocols
- OIP_Supplementary_Diagnostic_Self_Test

NOVEL CONTRIBUTIONS

1. Seven-gate epistemic architecture covering source attribution, evidence balance, uncertainty disclosure, causal reasoning, scope management, logical coherence, and representational discipline. Control mechanisms using Graduated Enforcement Architecture and json schemas
2. Epistemic Category Gates (ECG) for context-sensitive burden modulation across six question types (E1-E6)
3. Comprehensive evaluation methodology (OIP-TF) with numeric integrity scoring

SCOPE AND LIMITATIONS

This framework is presented as a theoretical and architectural specification rather than a validated deployment system. Key limitations explicitly acknowledged:

- No formal empirical validation studies have been conducted (although a small self-test has been provided)
- Framework relies to some degree on probabilistic LLM evaluation
- Claim segmentation introduces evaluator variance
- Token overhead limits high-volume deployment
- No reference implementation of orchestration layer provided

The framework is designed for high-stakes, low-volume contexts where epistemic safety is prioritized over efficiency or fluency. It is intended as decision-support infrastructure for expert human review, not autonomous compliance certification.

INDEPENDENCE STATEMENT

This research was conducted entirely independently by the author outside of professional employment. The framework has been developed without institutional affiliation, organizational funding, or access to proprietary systems or data.

While the author is employed in financial services, this work is not affiliated with, endorsed by, or deployed within any organization. Views expressed are solely those of the author.

SEEKING COLLABORATION

The author actively seeks academic collaboration to conduct empirical validation studies, particularly inter-rater reliability analysis and systematic evaluation across diverse high-stakes domains.

Potential collaborators interested in:

- Conducting validation studies

- Domain-specific adaptations (medical, legal, financial)
- Reference implementation development
- Integration with existing frameworks

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CITATION

If you use or reference this work, please cite as:

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VERSION HISTORY

v1.1 (January 2026) - Initial public release

- Complete framework specification
- Academic paper submission
- Comprehensive worked examples
- Explicit limitations analysis

ACKNOWLEDGMENTS

Developed independently with no institutional funding or organizational support.
Framework design informed by 20+ years professional experience in financial crime compliance, regulatory governance, and evidence evaluation.

CONTACT & UPDATES

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For updates, corrections, or additional materials, check the Zenodo record for new versions. Last updated: January 2026